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Review of “Opportunities and Strategies for Testing and Infusion of ISRU in the Evolvable Mars Campaign”

A manned Mars mission poses daunting logistical challenges. Included in the myriad of difficulties is how to best provide the necessary resources. In-Situ Resource Utilization (ISRU) is an option that lowers the launch mass which in turn decreases fuel costs and time spent in transit, increasing the efficiency of the expedition. Mueller et al. analyze the applications of ISRU systems and the benefits of an evolutionary strategy to develop and deploy ISRU technologies into the Evolvable Mars Campaign (EMC), both of which are necessary for a sustained human presence in deep space.

Integrating ISRU systems has practical applications for a manned Mars mission which can increase flexibility and decrease the expedition’s dependence on Earth, enhancing the survival capabilities of humans on other planets. For example, water can be acquired through ISRU by drilling into polar ice caps or the process of desorption by heating hydrated minerals found in regolith on the Martian surface (Mueller et al. 6). ISRU systems can also be used to produce hydrocarbon fuel for the Mars Ascent Vehicle (MAV) (Mueller et al. 2). In addition to sustaining human life, water can be electrolyzed into oxygen and hydrogen, and the latter can then be used in a Sabatier reaction to generate methane which is used along with oxygen to produce propellant for the MAV (Mueller et al. 5-6). Finally, the transit vehicle could capture and process asteroidal material to protect astronauts from radiation emitted by solar energetic particles in deep space (Mueller et al. 4). Reusing ISRU systems to continue these applications and the pioneering of efficient methods of resource extraction on an initial Mars expedition would further decrease the amount of materials needed for future missions, increasing the efficacy of future missions and enhancing the ability of humans to survive in an extraterrestrial environment (Mueller et al. 2).

According to Mueller et al., ISRU technology should be gradually developed and implemented into the EMC (Mueller et al. 2). The evolutionary process of developing and testing ISRU systems allows potential crew members to gain thorough knowledge about the environments they will encounter and experience using ISRU technologies (Mueller et al. 2). First, prospecting for resources is essential to understand their location, concentration, and form for extraction (Mueller et al. 3). To determine these characteristics of the Martian environment, regolith samples are heated to release volatile gases analyzed by a gas chromatograph or a mass spectrometer which can be accomplished by robotic missions (Mueller et al. 3). Next, in order to test the capabilities of ISRU technologies on a small scale, the EMC will need to utilize terrestrial test facilities, the ISS, and future lunar missions (Mueller et al. 3-4). To scale up the process of ISRU systems development, the EMC could utilize the Exploration Augmentation Module (EAM) as an external carrier platform because the hazardous nature of space resource processing will require that the processes occur exposed to space (Mueller et al. 4). The use of the EAM would allow the development of ISRU technology adapted to the microgravity, thermal characteristics, and vacuum of space (Mueller et al. 4). Finally, a crewed mission to a carbonaceous C-type asteroid redirected to cis-lunar space would allow astronauts to gain experience relying on ISRU systems, allowing them to develop problem solving skills necessary for survival (Mueller et al. 4). The mission could also provide the opportunity to test ISRU technology in an environment with similar characteristics to those encountered on a manned Mars mission (Mueller et al. 4). The experience with ISRU systems and knowledge of extraterrestrial environments gained during each aspect of the evolutionary development process will enhance the self-reliance of astronauts on deep space missions, allowing humans to sustain an interplanetary presence.

ISRU is appealing for a manned Mars mission because it lessens the logistical constraints that resource transportation imposes on the expedition. The combination of the applications of reusable ISRU technologies to lessen the logistical difficulties regarding resource transportation paired with the pioneering culture of self-reliance that results from an evolutionary approach to implementing ISRU systems results in the ability for humans to develop an interplanetary civilization.

Works Cited

- Mueller, Robert, et al. "Opportunities and Strategies for Testing and Infusion of ISRU in the Evolvable Mars Campaign." *American Institute of Aeronautics and Astronautics*, 31 Aug. 2015, pp. 1-7.
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